RV College of Engineering, Bengaluru – 560059

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ECO-SYNC TRANSIT AI

Agentic AI for Economic Dispatch and Anti-Bunching in Urban Transit

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Introduction

Public bus networks in Indian cities suffer from **bus bunching** — a dynamic instability where a minor delay causes one bus to accumulate more passengers, slow further, and cluster with trailing buses. On **Bengaluru's Outer Ring Road**, this collapses headway spacing and doubles commuter wait times. **ECO-SYNC Transit AI** replaces static timetables with an Agentic PPO Reinforcement Learning dispatcher optimising a multi-dimensional **Economic Reward Function**, explained in real-time by **Groq Llama-3.1**.

Problem Definition

1. Buses cluster, causing cascading gaps exceeding 20 minutes on Bengaluru ORR.
2. Static timetables cannot adapt to live congestion or passenger surges.
3. Existing RL controllers minimise headway variance but ignore economic costs.
4. Dispatch decisions are black-box, eroding driver and commuter trust.
5. No open-source system provides real-time dispatch *and* explainability together.

Objectives

1. Build a high-fidelity Gymnasium environment: 30-stop Bengaluru ORR circular route.
2. Design Economic Reward: R = Revenue – Wait_Cost – Fuel_Cost – Bunching_Penalty.
3. Train PPO agent to issue PROCEED / HOLD / SKIP actions preventing bunching cascades.
4. Integrate Groq Llama-3.1 to generate plain-language dispatch justifications in real-time.
5. Deploy React.js + FastAPI live dashboard with Leaflet maps and WebSocket telemetry.

Methodology

1	2	3	4	5
STATE EXTRACT	**ACTION SELECT**	**ECON REWARD**	**LLM REASON**	**WS STREAM**
S=[gap_ahead, gap_behind, queue_len]	PPO outputs PROCEED / HOLD / SKIP	R = Rev–WaitCost–FuelCost–Penalty	Groq Llama-3.1 explains action live	FastAPI 1Hz → React → Leaflet map

Economic Reward Function

R = Revenue – Wait_Cost – Fuel_Cost – Bunching_Penalty

Revenue = Rs.15 × boarded_passengers

Wait_Cost = passengers_waiting × Rs.1.67/min × delay × 10

Fuel_Cost = Rs.0.83/min × idle_minutes

Bunching_Penalty = -250 if gap_ahead <= 1.0 stop

Tools Used

- **Gymnasium / SB3** — PPO training environment
- **FastAPI + WebSocket** — 1Hz telemetry streaming
- **React.js + Vite** — Live dashboard frontend
- **Leaflet / react-leaflet** — Bengaluru ORR map layer
- **Groq Llama-3.1** — Real-time dispatch explanations
- **OSRM API** — Real road coordinate routing

Live Dashboard (Screenshot)

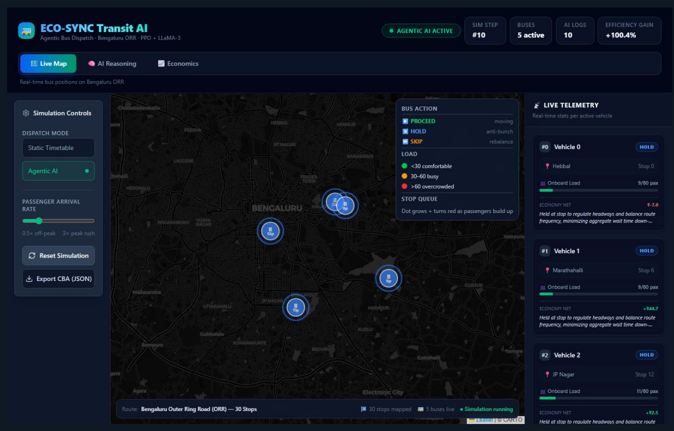
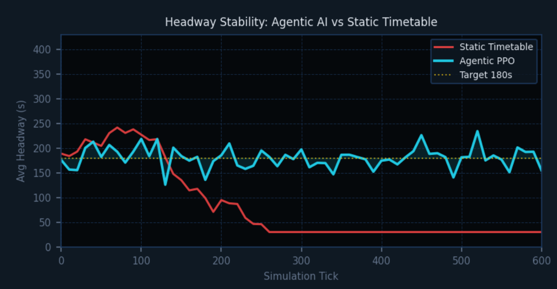
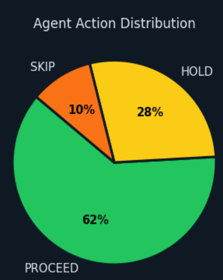


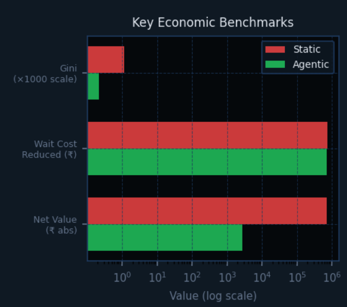
Fig 1: Real-time bus positions on Bengaluru ORR — Agentic AI Active, 5 buses live

+100.4%	97.3%	0.22	Rs.2,756
Net Efficiency Gain	Wait Cost Saved	Gini Equity Index	Net PPO Profit

Results & Discussion







Key Numerical Results

Metric	Static Timetable	Agentic PPO
Net Economic Value	-Rs.7,10,981	+Rs.2,756
Passenger Wait Cost	Rs.7,33,858	Rs.20,119 (-97.3%)
Fleet Fuel Cost	Rs.372.78	Rs.374.85 (+0.55%)
Ticket Revenue	Rs.23,250	Rs.23,250
Gini Equity Index	1.16 (Unfair)	0.22 (Fair)
Net Efficiency Gain	Baseline	+100.4%

Conclusions

ECO-SYNC Transit AI demonstrates that framing urban bus dispatch as an **economic optimisation problem** leads to dramatically superior outcomes. The PPO agent learned to HOLD and SKIP stops strategically, preventing bunching cascades and delivering a **Rs.7,13,739 reduction** in passenger wait costs and a **100.4% net efficiency gain** over the static baseline. Groq Llama-3.1 provides fully explainable, rider-facing justifications for every decision. The Gini equity index fell from 1.16 to 0.22, confirming uniform, fair service across all 30 ORR stops.

Outcome of the Work

A fully functional **Transit AI prototype**: (1) Custom Gymnasium environment with 30-stop Bengaluru ORR, Poisson arrivals, stochastic delays; (2) Trained PPO agent achieving headway stability across 250 Monte Carlo episodes; (3) FastAPI WebSocket backend at 1Hz; (4) React.js dashboard with Leaflet maps, live economics charts, LLM-generated explanations; (5) Research manuscript targeting **IEEE Trans. on Intelligent Transportation Systems** documenting the economic reward formulation and benchmarks.

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Mentor

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